

AZAM MOHAMED

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EDUCATION

University of California, Santa Cruz, Santa Cruz, CA

Sep 2023 – Aug 2026

B.S. Computer Engineering, Minors in Electrical Engineering and Computer Science

Relevant Coursework: CSE 134: Operating Systems, CSE 130: Computer Systems, CSE 240: Artificial Intelligence, Data Structures

WORK EXPERIENCE

UC Santa Cruz jLab (Josephson Lab), Undergraduate Researcher, ENTS Project, Santa Cruz, CA

Jul 2026 – Present

- Trace the full ENTS data path, firmware to transport to the DirtViz dashboard, to locate where field readings are lost or silently corrupted before display.
- Decode nanopb Protocol Buffer telemetry to verify packets survive transport intact, isolating failures to a specific stage rather than the pipeline at large.
- Build a repeatable validation procedure so any lab member can certify a node's data end to end, replacing ad hoc spot checks that let failures reach production dashboards.

UC Santa Cruz, Undergraduate Researcher, Independent Study (CSE 198), Santa Cruz, CA

Jun 2026 – Present

- Configuring and testing the department's PICO compute cluster for Kubernetes research workloads under Prof. Abel Souza.
- Evaluating Incus system containers and VMs on Linux to build reproducible, isolated experiment environments, prototyping container and VM provisioning workflows.

Santa Cruz Beach Boardwalk, Rides Operator, Santa Cruz, CA

Feb 2026 – Jun 2026

PROJECTS

Slurm Cluster Administration (UCSC HPC Club)

Linux, Slurm, BeeGFS, NFS, Rocky Linux

- Administer a six-node Slurm cluster (login controller, four compute nodes, and a routed GPU node) running Ubuntu over 1 GbE, including node state recovery through Slurm.
- Migrated the shared filesystem from a single-node NFS export to BeeGFS striped across unused disks on all four compute nodes, moving 333,879 files with ownership and permissions preserved and no downtime on /home.
- Wrote deployment scripts and an operational runbook covering verification, troubleshooting, and rollback, and ported cluster provisioning (repos, mDNS, munge, Slurm, users, NFS) to Rocky Linux.

Pintos Kernel Scheduler

C, OS Internals, Thread Scheduling

- Reworked Pintos thread sleeping to use an ordered sleep_list with a custom wakeup_less comparator, replacing busy-waiting with efficient timed wakeups inside the kernel.
- Woke sleeping threads from the 8254 timer interrupt handler by popping due threads from the sorted list front, disabling interrupts around insertion to avoid races with the ISR; passed all six alarm tests under QEMU.

Thread-Safe Binary Search Tree

C, pthreads, Mutexes, Valgrind

- Implemented a thread-safe binary search tree in C using fine-grained mutex locking to support concurrent insert, search, and delete operations across multiple threads.
- Hardened it against races and leaks using Valgrind and multi-threaded stress tests, reaching clean runs with zero reported memory errors under sustained contention.

Multi-Client TCP Chat Server

C, POSIX Sockets, select()

- Built a TCP chat server in C using select()-based I/O multiplexing to serve many clients concurrently on a single thread, handling connects, disconnects, broadcast, and clean session teardown without disrupting other users.

AI Search, Optimization, and Reasoning (CSE 240)

Python, python-chess, NumPy

- Implemented BFS, DFS, and A* with an admissible heuristic for knight pathfinding, then minimax with alpha-beta pruning and expectimax over chess positions, using move ordering and top-K pruning to cut the branching factor.
- Built local-search optimizers (hill climbing with restarts, simulated annealing, local beam search) and a min-conflicts CSP solver, plus Bayesian-network inference and a backward-chaining rule engine.

2x2 Matrix Solver ASIC (TinyTapeout, SKY130)

Verilog, cocotb, OpenLane, Fixed-Point

- Designed a sequential 2x2 linear-system solver ($Ax=b$, Q4.4 fixed point) in Verilog for the TinyTapeout SKY130 flow, fitting the datapath and a 12-state control FSM in a 1x1 tile targeting 50 MHz.
- Verified via cocotb through RTL and gate-level simulation, then hardened to GDS with the OpenLane toolchain.

SKILLS

Languages: C, C++, Python, MATLAB, Verilog, Assembly (x86/RISC-V)

Systems: Linux, Slurm, BeeGFS, NFS, Operating Systems, Concurrency, POSIX Threads, Sockets, select() Multiplexing

Tools: Git, GDB, Valgrind, Docker, NumPy, pandas, scikit-learn, cocotb, SQLite, WSL, SSH, Neovim, tmux